Research presentation

Explainable AI for Sand and Dust Storm Forecasting in Riyadh

Applying Machine Learning Algorithms to Improve the Accuracy of Medium- to Long-Term Sand and Dust Storm Forecasting

The problem: dust storms in Riyadh

Strong dust sources nearby

Riyadh lies close to the Ad-Dahna dune corridor and downwind of the Rub' al Khali. The shamal winds of March to July bring the strongest activity.

Forecasts lose accuracy fast

The raw CAMS forecast of Riyadh PM10 three days ahead has an RMSE of about 970 µg/m³ in the test period.

A warning gap

Planners need outlooks at 1, 3 and 7 days, both as PM10 values and as the probability of a severe event.

In the dataset, Riyadh PM10 averages 675 µg/m³ and peaks at 3,271 µg/m³.

How AI technology helps

Learns forecast errors

A model predicts the correction added to the raw NWP value: final forecast = NWP + learned residual.

Detects severe events

Classifiers turn 25 engineered features into a storm probability at +24h, +72h and +168h.

Quantifies uncertainty

Quantile models give a P10, P50 and P90 range instead of one number.

Explains itself

Explainable AI shows which inputs drove each forecast, so forecasters can check it against physics.

Project pipeline

1 Data

CAMS / ECMWF NWP products, 6 stations, hourly.

2 Exploration

Distributions, seasonal cycles, exceedance curves, correlation.

3 Feature engineering

41 raw columns grow to 99. 25 become model inputs.

4 Modelling

Logistic Regression, Random Forest, XGBoost, residual and quantile regressors.

5 Testing

Chronological split, ROC-AUC, PR-AUC, RMSE, PICP and skill scores.

6 XAI

Feature importance now, SHAP and permutation analysis next.

The data

Six stations

Riyadh Center is the urban receptor. Ad-Dahna, Nafud Al-Thuwairat and Rub' al Khali are dust sources. Al-Thumamah and Tuwayq Escarpment lie between.

Hourly NWP products

Copernicus CAMS / ECMWF forecasts from August 2022 to October 2026.

37,200 rows, 41 columns

Five NWP variables per station, time fields, and six target columns.

Data types and their meaning

Group	Variables	Type and unit
Time	year, month, day, hour	Integers taken from the timestamp
Particles	pm10, pm2.5	Continuous, µg/m³
Dust	dust_surface	Continuous, µg/m³
Optical	dust_aod_550nm	Continuous, unitless column dust depth
Radiation	uv_index	Continuous index
Target value	target_pm10_1d, 3d, 7d	Continuous PM10 at +24h, +72h, +168h
Target event	target_event_1d, 3d, 7d	Binary label, 1 for a severe event

Every station carries the five NWP variables, which makes 30 of the 41 columns.

An example of the data

Inputs: Ad-Dahna station, first three hours (4 August 2022)

time	pm10 (µg/m³)	pm2_5 (µg/m³)	dust_surface (µg/m³)	dust_aod_550nm	uv_index
2022-08-04 00:00	60.8	30.0	55.0	0.47	0.0
2022-08-04 01:00	58.2	29.3	51.0	0.46	0.0
2022-08-04 02:00	55.1	28.1	45.0	0.46	0.0

Targets on the same rows

time	target_pm10_1d	target_event_1d	target_pm10_3d	target_event_3d
2022-08-04 00:00	54.0	0	78.0	0
2022-08-04 01:00	55.5	0	92.2	0
2022-08-04 02:00	57.7	0	135.0	0

Feature engineering: turning raw fields into signals

Transport gradients

Source PM10 minus Riyadh PM10 shows dust flowing toward the city.

Source aggregates

Mean and max PM10 and mean AOD across the three source basins.

Optical ratios

PM10 to AOD, PM2.5 to PM10, and log transforms.

Temporal memory

Lags of 1 to 24 hours, rolling mean, std and max over 6 to 72 hours.

Cyclic time

Sine and cosine of hour and month, so 23:00 sits next to 00:00.

25 model inputs

The final set used by the classifiers and regressors.

Feature engineering: examples

Feature	Formula	Meaning
gradient_dahna_riyadh	PM10 Ad-Dahna – PM10 Riyadh	Dust pressure toward the city
surface_to_column_ratio	PM10 ÷ (AOD + 0.0001)	How much column dust reaches the ground
pm25_to_pm10_ratio	PM2.5 ÷ (PM10 + 0.0001)	Fine versus coarse dust
pm10_roll72h_max	Maximum of the past 72 hours	Recent peak, the dust memory
hour_sin, month_cos	sin or cos of 2π × time ÷ period	Smooth daily and seasonal cycles

XGBoost: the idea in general

1 A tree

A small decision tree splits the data with yes or no questions about the features.

2 Many trees

Trees are added one after another, each fitted to the mistakes left by the earlier ones.

3 Small steps

A learning rate scales every tree, so the model improves gradually.

4 Control

Depth limits and regularisation keep the model from memorising noise.

Prediction = tree 1 + η × tree 2 + η × tree 3 + …, where η is the learning rate.

XGBoost customised for this project

Setting	Value	Role in the project
Task	Storm yes or no	One model each for +24h, +72h, +168h
Inputs	25 engineered features	Standardised with training data only
n_estimators	120	Number of boosted trees
max_depth	6	Depth of each tree
learning_rate	0.08	Step size of each tree
scale_pos_weight	Negatives ÷ positives	Gives the rare storm class more weight
eval_metric	logloss	Rewards well-calibrated probabilities

Boosting in three roles

Storm classifier

XGBoost predicts the probability of a severe event at three lead times.

Bias corrector

A HistGradientBoosting regressor learns the NWP error: 150 iterations, learning rate 0.08, 31 leaves per tree.

Uncertainty bands

Quantile regressors with pinball loss at 0.10, 0.50 and 0.90 give the P10 to P90 range.

Testing design: a chronological split

70%

Training

26,040 samples to fit the models.

15%

Validation

5,580 samples for tuning and early stopping.

15%

Test

5,580 newest samples, used only for scoring.

Samples stay in time order so the model never trains on the future.

Testing parameters: storm classification

Metric	Question it answers	How to read it
ROC-AUC	Ranks storm hours above calm ones?	0.5 is chance, 1.0 is perfect
PR-AUC	Keeps precision as recall grows?	Compare with the 10% event rate
Precision	Share of alerts that were real	High means few false alarms
Recall	Share of real storms caught	High means few misses
F1-score	Balance of precision and recall	Harmonic mean of the two
Accuracy	Share of correct labels	Misleading when 90% of hours are calm
Brier score	Mean squared probability error	Lower is better

Testing parameters: bias correction and intervals

Metric	Meaning	Project target
RMSE	Typical error, large misses count most	Cut raw NWP error by 25% or more
MAE	Average absolute error	Lower than raw NWP
R^2	Share of variance explained	0.70 or higher
PICP	Share of observations inside P10 to P90	Nominal 80%, target 85% or more
MPIW	Average width of that interval	Narrow, but only with good coverage

Testing parameters: categorical skill scores

POD

Probability of detection

Hits ÷ (Hits + Misses). Target 80% or higher.

FAR

False alarm ratio

False alarms ÷ (Hits + False alarms). Target 25% or lower.

CSI

Critical success index

Hits ÷ (Hits + Misses + False alarms). Target 60% or higher.

Notebook 03 scores these at 150 µg/m³. That is far below Riyadh's mean of 675 µg/m³, so most hours likely exceed it and the reported POD of 99.87% says little. A higher threshold is needed.

Test results at +24h

Model	ROC-AUC	PR-AUC	Precision	Recall	F1
Logistic Regression	0.647	0.296	0.212	0.781	0.333
Random Forest	0.696	0.317	0.284	0.553	0.376
XGBoost	0.664	0.291	0.279	0.460	0.347

Logistic Regression catches the most storms but raises many false alarms. Random Forest balances best, and XGBoost is the model explained in this work.

Skill fades with lead time

Horizon	Best model	ROC-AUC	PR-AUC
1 day (+24h)	Random Forest	0.696	0.317
3 days (+72h)	XGBoost	0.584	0.224
7 days (+168h)	Random Forest	0.595	0.205

The event rate is about 10%. The 3-day bias corrector cuts RMSE from 969.9 to 773.3 µg/m³ (20.3%), short of the 25% target, and R^2 stays at −0.08.

Explainable AI: why it matters for early warning

Trust

A forecaster needs the reason behind an alert, not only a probability.

Physical check

Does the model use dust transport, season and recent dust the way atmospheric physics says it should?

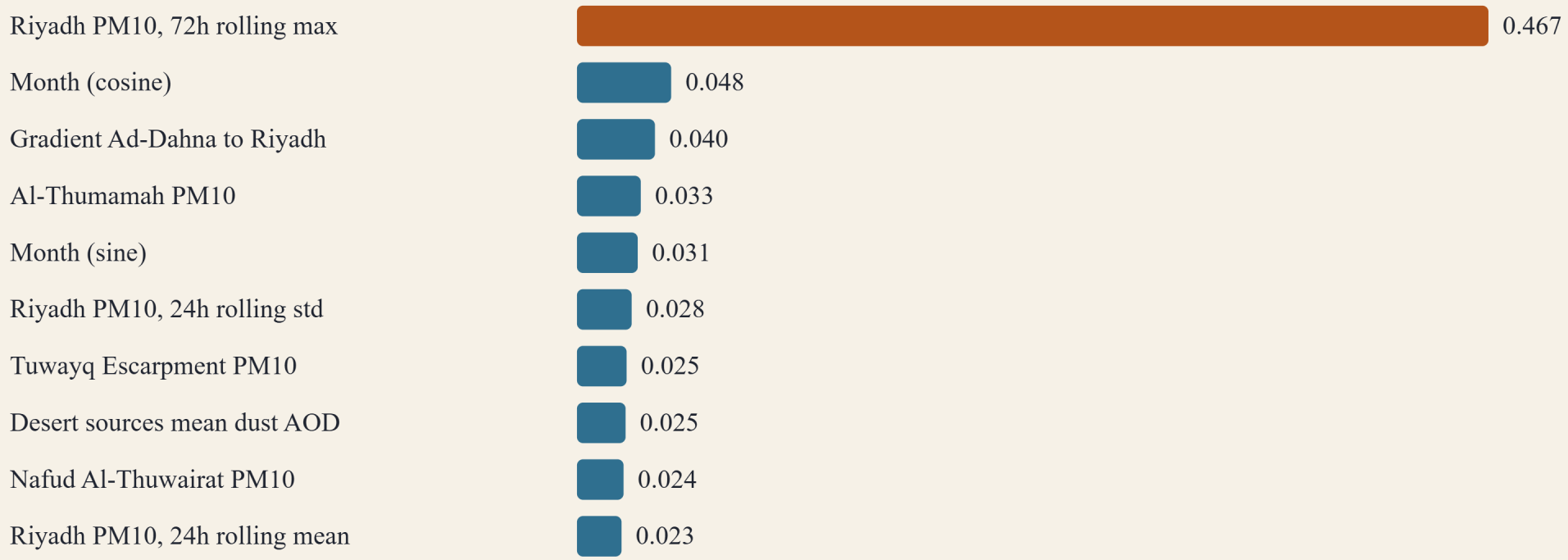
Diagnosis

Explanations show which drivers fade at 3 to 7 days, where skill drops.

The XAI toolbox for this study

Method	Question it answers	Status
Gain importance (XGBoost)	Which features drive the splits?	Done, notebook 04
Quantile P10 / P50 / P90	How uncertain is the forecast?	Done, notebook 03
Linear coefficients	Which direction does each input push risk?	Model trained, not yet read
Permutation importance	Which features does skill depend on?	Proposed
SHAP values	How much, in which direction, for each day?	Proposed
PDP / ICE curves	How does risk change as one feature varies?	Proposed

XGBoost feature importance, 3-day storm classification



Top 10 of 25 features by gain importance. Source: notebook 04.

Persistence leads, then season and upstream transport

51.8%

Recent dust memory

72h max, 24h std and 24h mean of Riyadh PM10.

14.7%

Upstream transport and optics

Ad-Dahna gradient, Al-Thumamah, Tuwayq, Nafud and dust AOD.

7.9%

Seasonal cycle

Month sine and cosine, the spring and summer shamal season.

Shares of total importance, summed over the top 10 features.

Uncertainty is part of the explanation

66.34%

Interval coverage (PICP)

The P10 to P90 band should cover 80% of observations. It covers about two thirds.

1,404 µg/m³

Mean interval width

Wide bands, yet still too narrow: the 3-day model is overconfident.

Any explanation of a 3-day forecast should travel with its interval. Source: notebook 03.

Reading the explanations with care

Persistence dominance

A 47% share for one 72h feature suggests the model mostly extrapolates recent dust. Compare it against a persistence baseline.

Gain importance limits

It shows no direction, and credit is split among correlated lags, rolling statistics and station PM10.

Skill scores to verify

Notebook 03 reports POD 99.87% and CSI 96.57% next to R^2 of −0.08, because the 150 µg/m³ threshold is easy to exceed.

Station risk table

Notebook 04 shows the same risk set (16.8%, 0.5%, 7.4%) for all six stations because one feature vector was reused.

Next steps for the XAI analysis

SHAP

TreeExplainer on the 3-day XGBoost: global summary, dependence plots for rolling max and month.

Permutation

Chronological test set, grouped by physical family: memory, transport, season.

Case studies

SHAP waterfalls for real storm episodes, for example spring 2023.

By horizon

Compare drivers at 1, 3 and 7 days to see which fade.

Conclusions

What drives forecasts

Recent dust memory, then upstream transport and the seasonal cycle.

How far to trust them

Skill is modest beyond 1 day and the intervals under-cover.

What comes next

SHAP and permutation analysis, and a higher storm threshold.